

Exercise set 3

Textbook: Exercises 3.3, 3.5, 3.7, 3.17

Problem 1:

Consider a system consisting of 10 independent components with corresponding probabilities that they are working: $p_1 = 0.7$, $p_2 = p_3 = p_4 = 0.95$, $p_5 = p_6 = 0.99$, $p_7 = p_8 = p_9 = 0.92$ and $p_{10} = 0.7$. Find the probability analytically that the system is working under the following set of conditions:

- a) The system works if and only if components 1 and 2 both work.
- b) The system works if and only if at least one of components 1 or 2 works.
- c) The system works if and only if both component 1 and component 2 and at least one of components 3 or 4 work.
- d)^R Find the probabilities under the conditions a) to c) by simulation using 10 000 repetitions.
In addition, find the probability that the system is working if and only if at least 7 of the 10 components work.

Problem 2:

In a specific location the probability of finding oil is considered to be 0.4, and if there is oil the uncertainty related to the size of the oil reservoir is described by a triangle distribution, where $x_{\min} = 2$, $x_{\text{most probable}} = 6$ and $x_{\max} = 10$.

- a)^R By simulation find an estimate of the probability that $P(X \geq 8)$, both unconditionally and conditionally on there being a find.

In a region there are 10 potential oil reservoir locations. In the table below the resource distributions (for X_i , $i = 1, \dots, 10$) for each of these locations are listed. Here p_i is the probability of a find and min, mode (most probable value) and max are referring to a triangle distribution.

Number	p_i	min	mode	max
1	0.8	2	4	6
2	0.3	3	7	11
3	0.6	2	4	6
4	0.6	1	5	9
5	0.5	8	9	10
6	0.9	5	7	9
7	0.5	2	4	6
8	0.8	3	4	5
9	0.4	8	10	12
10	0.4	3	5	7

- b)^R By simulation find the risk-weighted (i.e. take the probability of a find into account) and the non risk-weighted expectation (conditionally on there being a find) for the total resource $R = \sum_{i=1}^{10} X_i$.
- c) Check the answers in b) by analytic computations.
- d)^R By simulation find an estimate of the accompanying standard deviation for the two expectations.
Find how many replications are necessary to be 95% certain that the errors in the estimates of the means are not exceeding 0.2.

Problem 3:

The gross economic income from two investments are denoted X and Y , and these are normally distributed variables with $E(X) = E(Y) = 1$, $\text{Var}(X) = 1$ and $\text{Var}(Y) = 3$.

- a) When the correlation $\rho(X, Y) = 0.7$, specify the covariance matrix for the bivariate normal variable $[X, Y]^T$.
- b)^R Use appropriate functions in the `mvtnorm` library as help to simulate the probability $P(X + Y \geq 3)$.
- c)^R Simulate the probability $P(X + Y \geq 3)$ when $\rho(X, Y) = 0$.
- d) Try to verify the results in b) and c) by analytical calculations.